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Credit - Drishti Ias

Biomarkers (Chemical Signatures) and Early Disease Detection

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Source: TH

Why in News?

Scientists have emphasised the growing potential of **chemical signatures** (commonly known as **biomarkers**) in revolutionising early disease diagnosis, particularly for neurological conditions like **Alzheimer 's and Parkinson's .**

- However, despite their promise, integrating these diagnostic tools into routine clinical practice remains hindered by significant hurdles related to accuracy, cost, and accessibility.

What are Chemical Signatures (Biomarkers)?

- **About:** Chemical Signature is a measurable **biological indicator** found in bodily fluids (like blood or cerebrospinal fluid) consisting of a unique pattern of specific proteins, molecules, elements, isotopes, or by-products.
- **Function:** It objectively signifies what is happening inside the body, indicating a normal biological process, a pathogenic disease state, or a response to medical treatments.
- **Working of Chemical Signatures:** When a disease develops, it triggers changes at the **molecular level** long before physical symptoms appear.
  - By measuring the presence, absence, or altered behaviour of specific proteins, doctors can identify underlying physiological shifts.
- **Detection Instruments:** Advanced analytical tools like **spectrometers** (which measure molecules by mass or light absorption) and **chromatographs** (which separate mixture components) are used to detect these minute chemical patterns.
- **State of Biomarkers in India:**
  - **GenomeIndia Project:** Launched in January 2020 and funded by the Department of Biotechnology (DBT), the project has completed the **whole-genome sequencing of over 10,000 individuals** representing diverse Indian populations.
    - As one of the **world's largest genetic databases** , **Genome India Project** serves as a critical foundation for discovering population-specific biomarkers and genetic risk factors tailored to South Asians.
  - **BHARAT Study:** The Indian Institute of Science (IISc), under its Longevity India Programme, has

launched the **BHARAT (Biomarkers of Healthy Aging, Resilience, Adversity, and Transitions) Study** to establish a "Bharat Baseline" tailored to the Indian population, reducing reliance on Western-derived biomarkers.

- Since ageing is influenced by **molecular, cellular, environmental and socioeconomic factors, chronological age** may not accurately reflect biological age. The study aims to develop biomarkers that better capture ageing **patterns among Indians.**

Significance of Biomarkers in Healthcare

- **Early and Pre-symptomatic Detection:** Biomarkers allow physicians to transition from late-stage symptomatic management to early molecular triage, detecting conditions years before cognitive or physical decline begins.
- **Non-Invasive Diagnostics:** Blood-based biomarker assays offer a cheaper, safer, and less painful alternative to invasive procedures like lumbar punctures (spinal taps) or expensive nuclear **PET (Positron Emission Tomography) scans** .
- **Targeted Therapies:** By differentiating between diseases with overlapping symptoms (e.g., differentiating types of dementia), biomarkers ensure patients receive the correct targeted therapies rather than broad-spectrum symptom suppressants.
  - India's precision medicine market is projected to grow at a CAGR of 16%, exceeding USD 5 billion by 2030. Biomarkers can accelerate this growth by personalised treatments and more targeted therapies.
- **Tracking Disease Progression:** Serial blood tests allow doctors to monitor how well a patient is responding to experimental drugs or ongoing treatments dynamically over time.

Common Biomarkers and Linked Diseases

Biomarker	Associated Disease	Clinical Indication
Tau (p-tau217) & Amyloid-beta	Alzheimer's Disease	Tracks the buildup of abnormal plaques and neurofibrillary tangles in the brain.
Alpha-synuclein	Parkinson's Disease	Detects misfolded protein clumps linked to the loss of nerve cells that control movement.
Neurofilament Light (NfL)	Multiple Neurological Disorders	Acts as a non-specific marker indicating nerve cell damage and neuroaxonal injury.
Neuronal Surface Antibodies	Autoimmune Encephalitis	Indicates the immune system mistakenly attacking brain proteins, aiding in reversible neuro-inflammatory treatment.

Major Global Genome Sequencing Projects

Project	Country/Region	Objective
All of Us Research Program	**USA**	Advance personalized medicine using genomic and health data.
Human Microbiome Project	**USA**	Study human-associated microbes and their impact on health.
100,000 Genomes Project	**UK**	Diagnose rare diseases and cancers through whole-genome sequencing.
1+ Million Genomes (1+MG)	**European Union**	Enable secure genomic data sharing for research.
Emirati Genome Program	**UAE**	Improve preventive and personalised healthcare.

What are the Challenges in Testing and Diagnosis through Biomarkers?

- **Accuracy and Specificity Issues:** Many biomarkers lack absolute specificity. For instance, the **Neurofilament Light (NfL) protein** indicates nerve damage but cannot pinpoint the exact disease causing it.
  - Furthermore, factors like age, kidney function, and other concurrent medications can alter biomarker levels, leading to a risk of false-positive or false-negative results.
- **High Costs and Lack of Infrastructure:** Specialised tests requiring advanced **immunoassays are expensive and largely confined to major metropolitan centres.**
  - District and rural hospitals often lack the automated platforms required for these tests, resulting in severe diagnostic deficits in rural India.
- **Lack of Uniform Standards:** There is a lack of harmonisation across testing laboratories.
  - Variations in laboratory methods and equipment can affect the consistency and interpretation of biomarker test results.
- **Ethical and Socio-economic Concerns:** Identifying preclinical diseases in asymptomatic individuals creates "patients-in-waiting," triggering severe psychological anxiety.
  - Additionally, a positive biomarker test on medical records could expose individuals to socio-economic discrimination, such as denial of life insurance or workplace bias.

Way Forward

- **Promoting Indigenous R&D:** The government and public sectors must support domestic biotechnology firms to engineer affordable, locally manufactured diagnostic reagents. This will reduce

dependency on costly imported testing platforms and democratise access.

- **Standardisation of Protocols:** The **Indian Council of Medical Research (ICMR)** and the **Central Drugs Standard Control Organisation (CDSCO)** should establish unified clinical guidelines and standard operating procedures (SOPs) for biomarker testing to ensure uniform laboratory interpretation.
- **Ethical Safeguards:** Clear legal frameworks must be established regarding genetic and biomarker data privacy to prevent insurance and employment discrimination against asymptomatic 'at-risk' individuals.
- **Capacity Building:** Expanding diagnostic infrastructure at the district hospital level and training medical professionals in biomarker interpretation is critical for integrating these tools into mainstream public health planning.

Frequently Asked Questions (FAQs)

1. What are chemical biomarkers?
Measurable biological indicators used for disease detection, diagnosis and monitoring.

2. Which biomarkers are linked to Alzheimer’s and Parkinson’s?
Tau and Amyloid-beta are linked to Alzheimer’s, while Alpha-synuclein is linked to Parkinson’s.

3. What are the major challenges in biomarker testing?
High costs, accuracy issues, inadequate infrastructure and data privacy risks.

4. What are the roles of ICMR and CDSCO?
ICMR supports biomedical research, while CDSCO regulates drugs and medical devices.

5. How can India improve biomarker-based diagnosis?
Through indigenous R&D, standardised testing, affordable diagnostics and stronger rural healthcare infrastructure.

[Watch Video on YouTube: ▶ **https://www.youtube.com/embed/9urfvJDk7YM**]

UPSC Civil Services Examination, Previous Year Question (PYQ)

Prelims

Q. Which one of the following statements best describes the role of B cells and T cells in the human body? (2022)

(a) They protect the environmental allergens. body

(b) They alleviate the body's pain and inflammation.

(c) They act as immunosuppressants in the body.

(d) They protect the body from diseases caused by pathogens.

Ans: (d)

Q. With reference to agriculture in India, how can the technique of 'genome sequencing', often seen in the news, be used in the immediate future? (2017)

1. Genome sequencing can be used to identify genetic markers for disease resistance and drought tolerance in various crop plants.
2. This technique helps in reducing the time required to develop new varieties of crop plants.
3. It can be used to decipher the host-pathogen relationships in crops.

Select the correct answer using the code given below:

(a) 1 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

Ans: (d)

Mains

Q. Why is there so much activity in the field of biotechnology in our country? How has this activity benefitted the field of biopharma? **(2018)**

Upgraded Agri Welfare Portal Launched for Farmers

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Source: PIB

The **Ministry of Agriculture and Farmers Welfare launched the upgraded Agri Welfare Portal** to provide farmers with single-window, mobile-based access to government schemes, financial assistance and agricultural information.

- **About:** It is a **single-window digital platform** providing farmers access to **government schemes, financial assistance and reliable agricultural information** through mobile phones.
- **Single-Window Access:** The portal brings information and services related to **schemes, subsidies, rules and digital services** onto one platform, reducing the need to visit multiple offices or websites.
- **Key Schemes:** It provides access to major initiatives including **PM-KISAN** , **AgriStack** **and** **Pradhan Mantri Fasal Bima Yojana (PMFBY)** through a common interface.
- **Transparency and DBT: Online eligibility checks, direct applications and** **Direct Benefit Transfer (DBT)** aim to simplify processes, improve transparency and reduce intermediaries.
- **Farmer-Centric Design:** The portal is **lightweight, mobile-friendly and accessible on basic smartphones** , even in areas with weak internet connectivity, providing information on **weather, scientific farming practices, seeds, fertilisers and market prices** .
- **Agricultural Ecosystem:** It also provides a common digital platform connecting **Farmer Producer Organisations, Primary Agricultural Credit Societies, scientists, researchers and startups** .
- **Capacity Building:** **Krishi Vigyan Kendras** will conduct training campaigns to help farmers, particularly small and marginal farmers, effectively use the portal and other digital tools.
- **Digital Standards:** The portal has been developed in accordance with the **Digital Brand Identity Manual 3.0** and **Guidelines for Indian Government Websites 3.0** , focusing on security, accessibility and usability.

Read More: **Kisan Sarathi Platform**